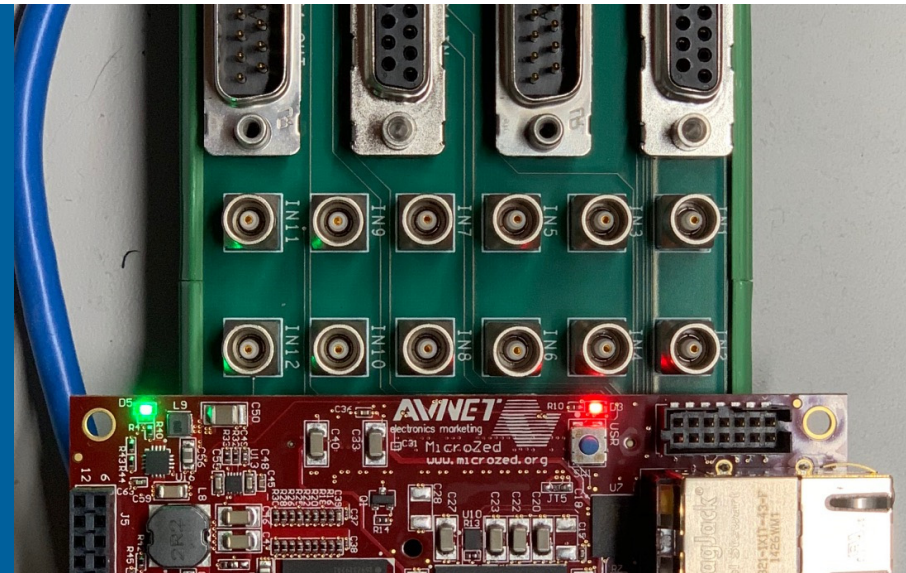


APS-U BEAMLINE CONTROL UPGRADE STRATEGY



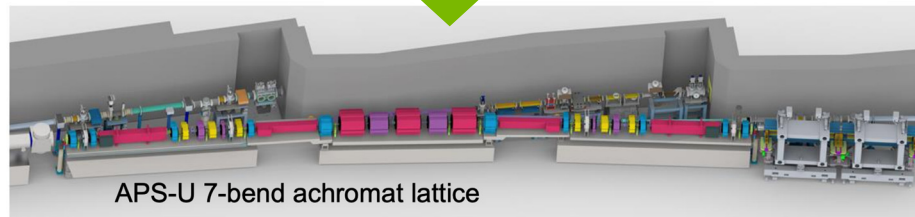
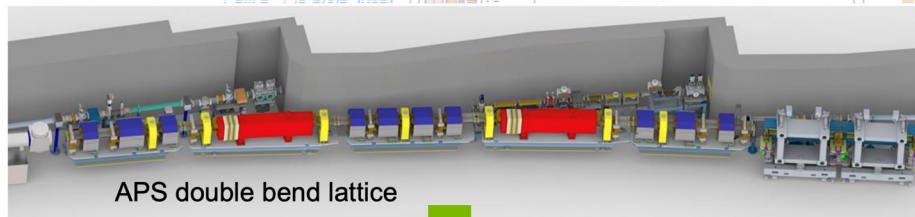
JOE SULLIVAN

Group Leader for APS Beamline Controls
X-Ray Science Division (XSD)

APS UPGRADE

APS-U is here!

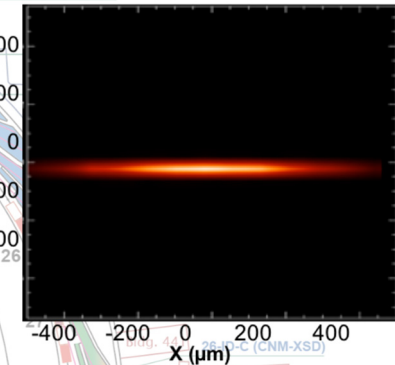
- The APS-U storage ring will feature a multi-bend achromat lattice, with many more bending magnets and magnet-focusing cells than the present machine, resulting in brighter X-ray production.
- One year dark period for storage ring rebuild starting ~ June 2022



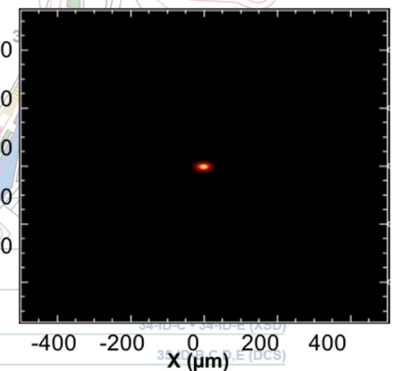
3000 pm-rad
emittance

42 pm-rad
emittance

APS Today



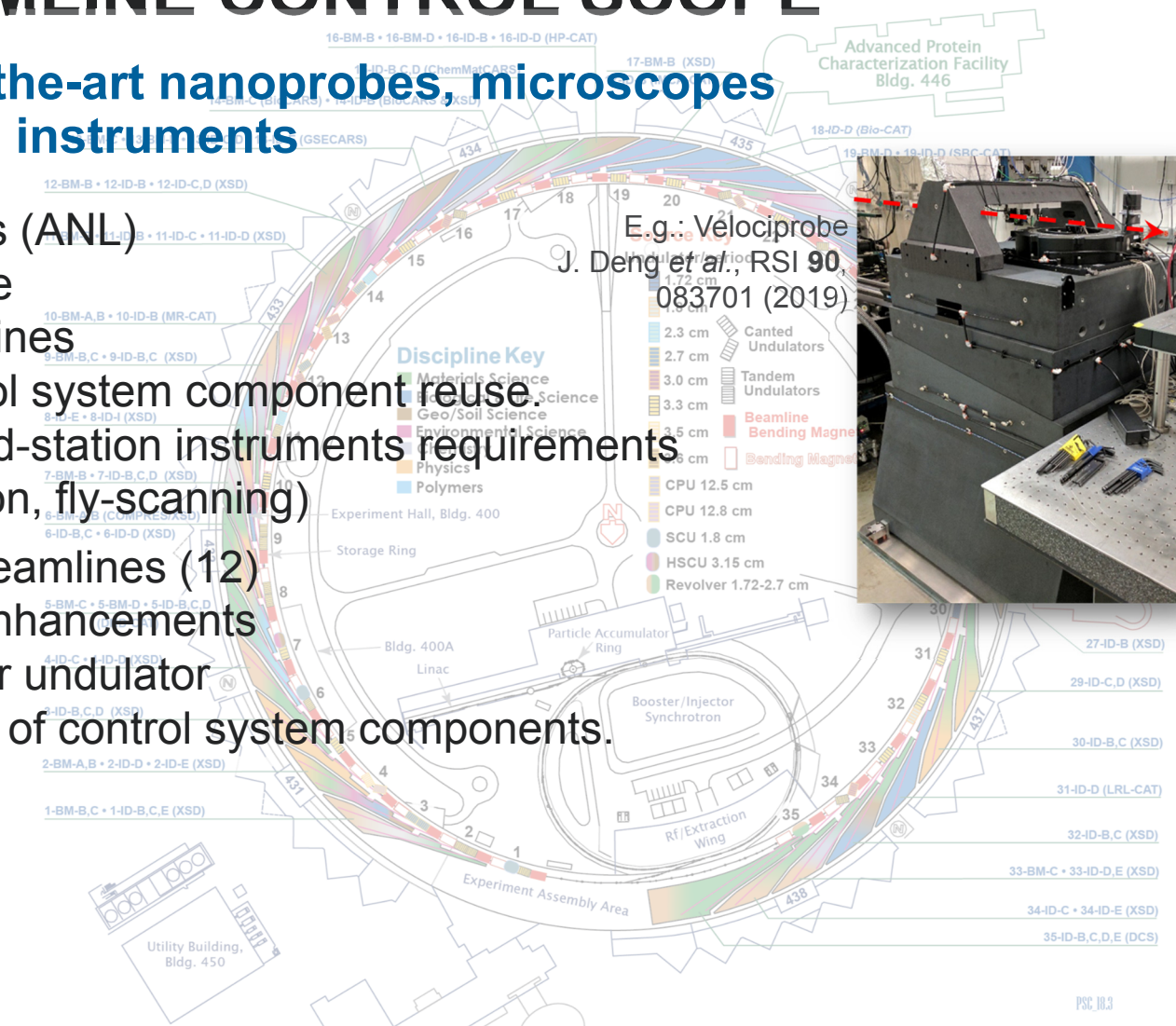
APS Upgrade



APS-U BEAMLINE CONTROL SCOPE

Suite of state-of-the-art nanoprobes, microscopes and tomographic instruments

- Feature Beamlines (ANL)
 - 1 new beamline
 - 7 rebuilt beamlines
 - Some control system component reuse
 - Demanding end-station instruments requirements (Nano-resolution, fly-scanning)
- Remaining XSD beamlines (12)
 - performance enhancements
 - optics and/or undulator
 - majority re-use of control system components.



BEAMLINE CONTROLS PHILOSOPHY

XSD Beamline Controls

- Operability
 - Performance meets requirements
 - Safety
 - Reliable



- Maintainability
 - Allows for **Continuous Improvement and Innovation.**
 - Field serviceable
 - Components can be replaced without disassembly (rewiring).
- Reusability
 - Toolkit design approach (synApps)

BEAMLINE CONTROLS SOFTWARE FRAMEWORK

EPICS + synApps + Bluesky



- EPICS V7
 - Combines Channel Access/PV with PVAccess development
 - Maintain existing CA/PV IOC and client applications and technologies
 - PVAccess - innovations in database and client/server technologies
- synApps (<https://github.com/epics-modules>)
 - EPICS beamline applications packaged for standard, convenient and rapid builds and deployment.

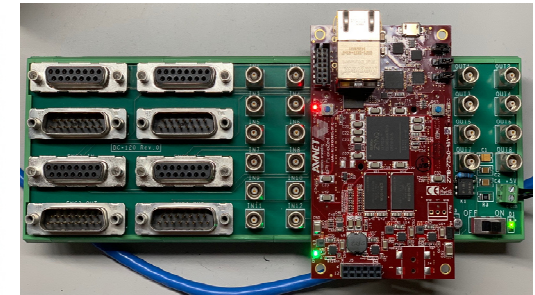
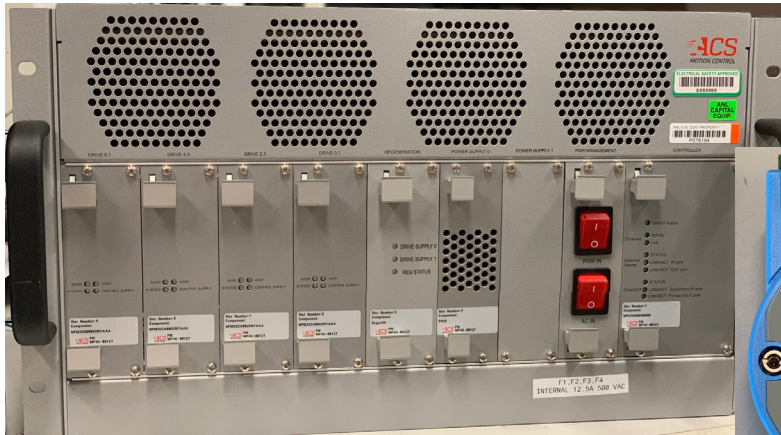


- Bluesky
 - Python based experimental control and data collection technology.
 - Collaboration effort across DOE lightsources (APS, NSLS-II, LCLS-II)

BEAMLINE CONTROL SYSTEM PLATFORMS

End-station instrumentation with advanced coordination and synchronization requirements.

- EPICS IOC's
 - **Linux** – Motor Controllers, I/O, Serial, Networked Devices, Cameras/Detectors
 - **Windows** (*when required*) – Detectors
 - **MicroZed** (FPGA + ARM) – softGlue, Scalers, Time/Position based triggering and data acquisition.



END-STATION / INSTRUMENT CONTROL SYSTEMS STRATEGY

softGlueZynq - field programmable FPGA applications

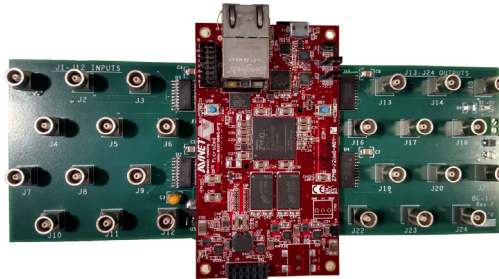
<https://github.com/epics-modules/softGlueZynq>

- Hardware:
 - Combined FPGA and an ARM processor (Xilinx Zynq chip) pre-packaged on an engineered board (microZed - AVNET)
 - Special use interface boards

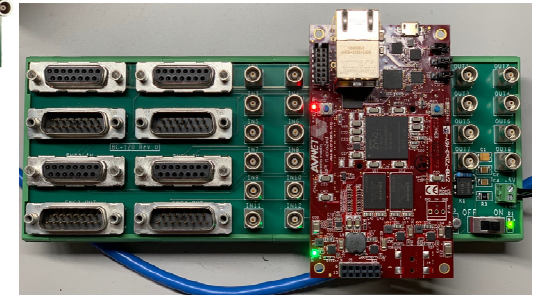
16 Channel Scaler (~40Mhz) +
Triggering



Triggering / Timing



Interferometer Recorder +
Triggering



- Integration:
 - Linux IOC (ARM Processor)
 - softGlueZynq: FPGA logic functions tightly coupled to EPICS (softGlue)

END-STATION / INSTRUMENT CONTROL SYSTEMS STRATEGY (CONT)

*“XSD/BCDA APS-U End-station/Instrument Control System Strategy”
(T. Mooney, K. Goetze)*

Risk Mitigation - softGlueZynq

■ Hardware:

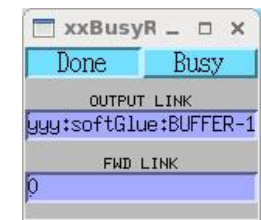
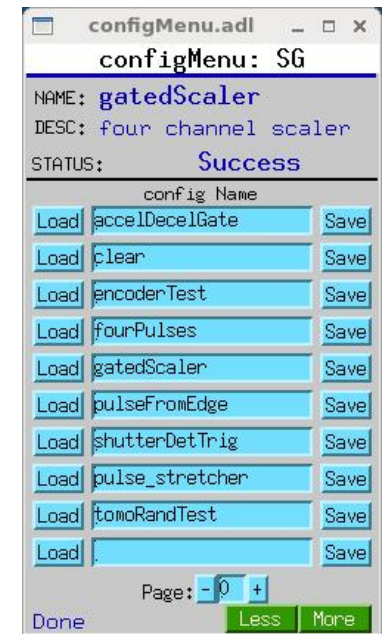
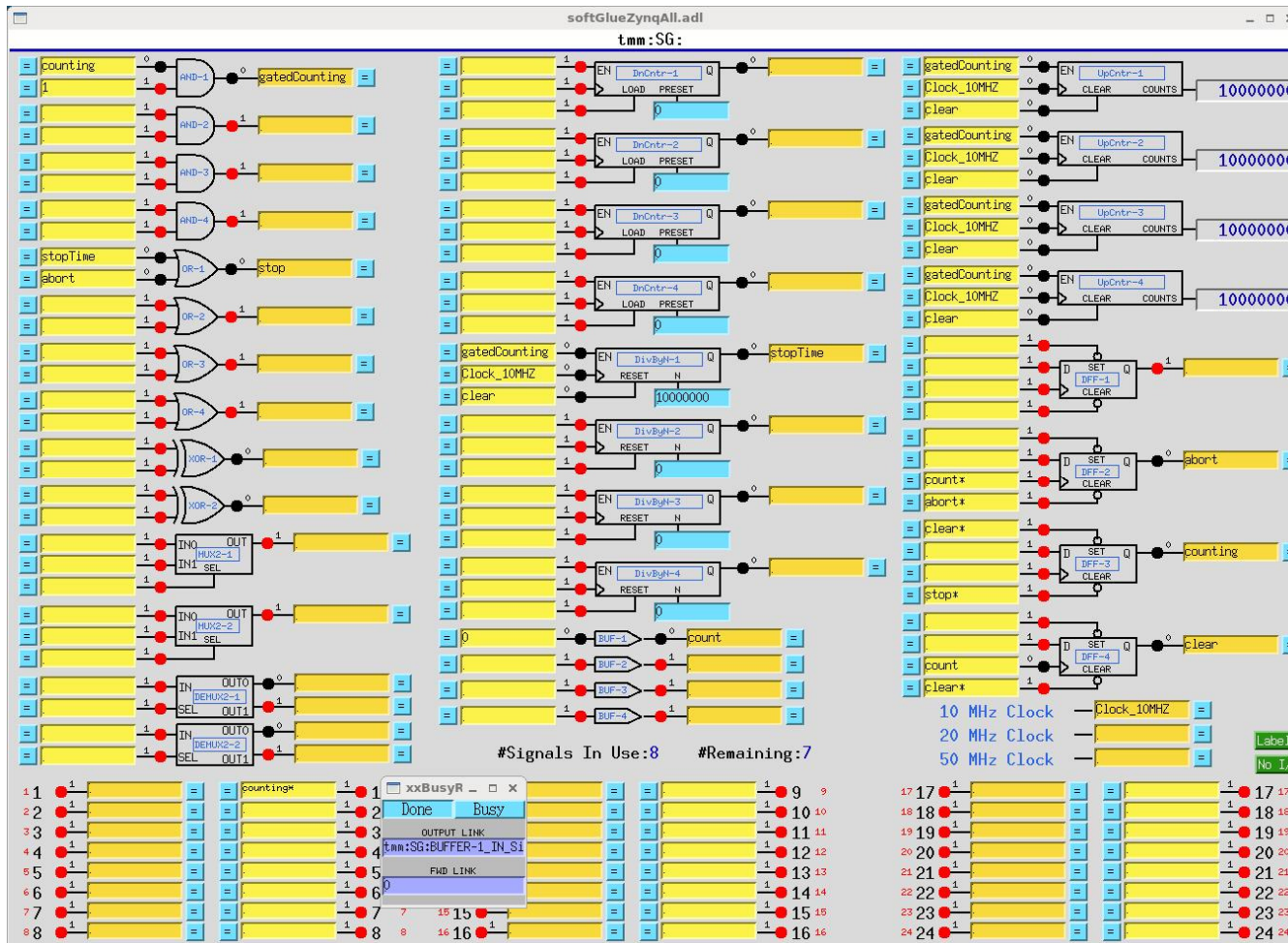
- Xilinx Zynq FPGA chip combined the AVNET ‘Zed’ electronics platforms are popular within the APS (ASD/Controls, XSD/Detector, DCS) and with other facilities (ie. PandABox – SOLEIL & Diamond)
- Industry standard FPGA programming environment (Verilog) enables portability to updated platforms.

■ Software

- EPICS IOC builds and deployment follows XSD standard softIOC process
- Extension of successful softGlue model (30 installed instances)
- Requirements and solutions are found and tested at running instruments (Bionanoprobe and >12 other instances deployed)

END-STATION / INSTRUMENT CONTROL SYSTEMS STRATEGY

softGlueZynq - simple digital electronic circuits created by writing to EPICS process variables (PVs).



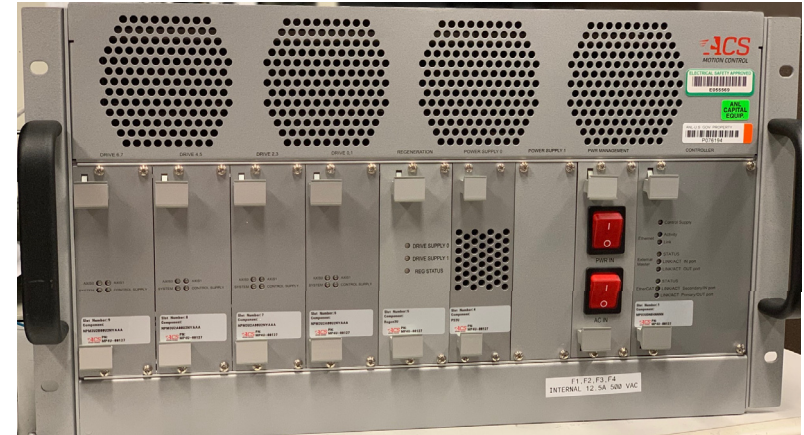
"T. Mooney, K. Goetze – "EPICS and Custom Beamline Electronics on the MicroZed", EPICS Collaboration Meeting, 2018

END-STATION / INSTRUMENT CONTROL SYSTEMS STRATEGY (CONT)

Advanced motion controls – realtime synchronization across devices

MP4U 8-Axis Drive Module (3 to 13 Amps)

ACS Motion Control
SPiiPlusEC (EtherCat)



- Programmable motion controller
 - Custom application runs at servo loop speeds (20kHz)
- Tailored motion solutions (pre-built chassis if needed)
 - Motor: Servo, stepper and piezo
 - Feedback: Quadrature, Absolute, Interferometer
- Position/vector trigger (18MHz Quadrature/4kHz Absolute)
- Axis synchronization and remote I/O over deterministic network
 - EtherCAT – 2kHz
- EPICS Support – `<synApps>/support/motor/motorApp/AcsTech80`

BEAMLINE CONTROL SYSTEM PLATFORMS

Photon Delivery - high motor count hutches (>40)

EPICS IOC (VME64x 16slot chassis - Vector)

- MVME SBC running commercial RTOS (vxWorks)
- OMA MAXv motor controller (V2 available year-end 2019)
 - 8 axis stepper/servo
 - quadrature and absolute encoder (BiSS)
 - no-wiring installation

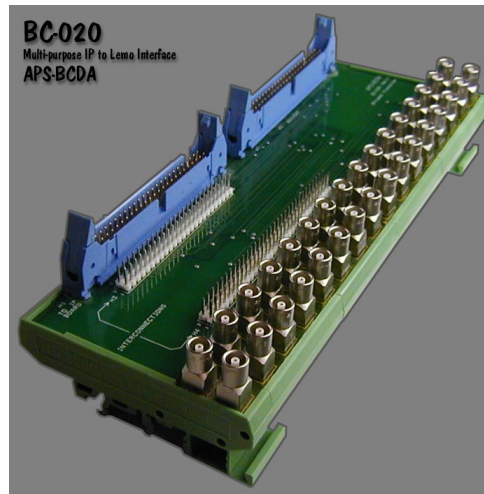


BEAMLINE CONTROL SYSTEM PLATFORMS (CONT)

Also VME gives additional flexibility to the control system

VME BASED I/O Boards

- Existing I/O solutions (IndustryPack) can still be used.
- Support popular scientific products (CAEN, Struck,...)



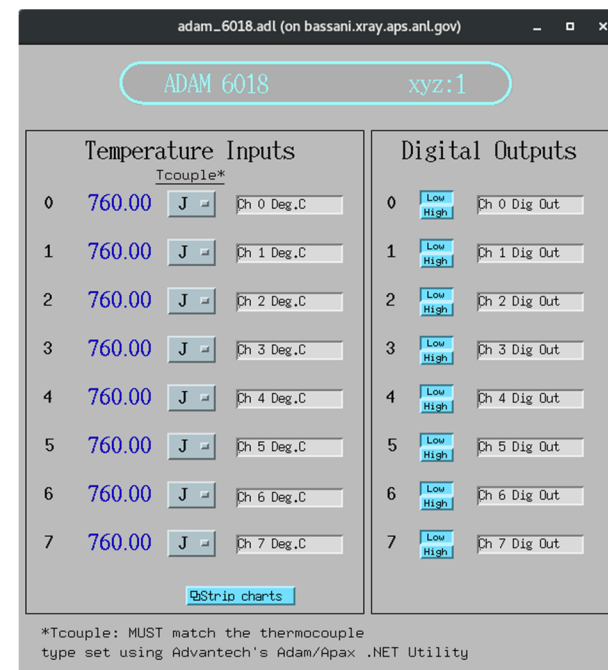
INTERNET OF THINGS

Networked I/O Devices

Selection Criteria

- Industrial customer base
 - Quality
 - Long life cycle
- Modbus interface
 - No driver development required
 - Asyn modbus Support

(<https://cars9.uchicago.edu/software/epics/modbusDoc.html>)
- Readily available (in-stock)



Advantech ADAM 6000 Series
Analog and Digital I/O, Relays,....

CONTROLS NETWORK

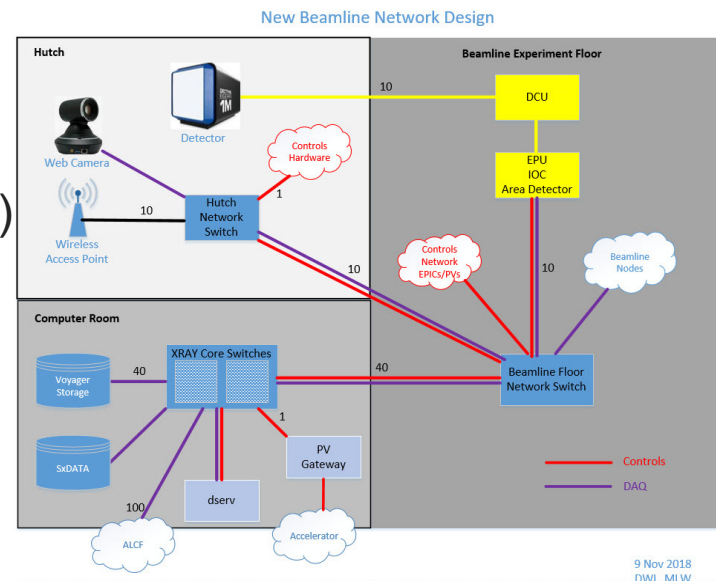
The network is the fieldbus (Internet of Things)

- Dedicate *local* network for beamline control system

- Isolate device and EPICS traffic from LAN
- Increased security
- ANL and DOE cyber requirements are more compatible with a controls networks (SCADA)

- Increase access to network ports

- Managed network switches in the hutch
 - Can be configured for High Performance, LAN or *local*
- Mountable on the instruments



SUMMARY

- Suite of state-of-the-art nanoprobes, microscopes and tomographic instruments requires control system innovations.
- Control system choices must allow for Continuous Improvement and Innovation
- Existing beamlines and instruments will continue to use existing controls
- EPICS + synApps + Bluesky (software framework)
- MicroZed FPGA platform with softGlue provides solutions for current instruments and is also provides an R&D platform for APS-U challenges.
- Motion system capable of synchronizing remote instruments (w/convenient packaging)
 - ACS Motion Systems is the current candidate.
- VME64 is a viable platform 'slow' control system motors and I/O.
- Networked I/O solutions (Modbus preferred)
- Secure Network Design (Local network for controls traffic)